

Unit 17b: Water Storage — Lakes, Groundwater & Aquifers

Reading: pages 254–259 · 20 points · 15 minutes · Closed book

Name _____

Date _____

Score / 20

Answer every question. For multiple choice, circle the single best answer. Write short answers in complete sentences.

Part A · Multiple Choice

6 questions × 2 points = 12 points

1. To be a good aquifer, a rock body needs three things:
 - a) hardness, density, and depth
 - b) salinity, temperature, and pressure
 - c) limestone, sandstone, and shale layers
 - d) porosity, permeability, and natural recharge by precipitation
2. The upper surface reached by groundwater within an aquifer is called
 - a) the water table
 - b) the recharge zone
 - c) the impermeable base
 - d) the aquifer
3. The Ogallala Aquifer supplies about what share of all irrigation water in the United States?
 - a) About one-tenth
 - b) About one-third
 - c) About one-half
 - d) About three-quarters
4. How much more water is withdrawn from the Ogallala Aquifer each year than is naturally replenished?
 - a) About four times as much
 - b) About eight times as much
 - c) About twenty times as much
 - d) About twice as much
5. The deep-water zone of a lake, which receives little or no sunlight, is also called the
 - a) littoral zone
 - b) limnetic zone
 - c) aphotic or profundal zone
 - d) photic zone

6. When coastal aquifers are overused, a common consequence is
- a) saltwater incursion into the aquifer
 - b) a sharp increase in local rainfall
 - c) a rise in the water table
 - d) the formation of new springs

Part B · True or False

2 questions × 1 point = 2 points

7. More than 60% of all the world's lakes are in Canada.

TRUE / FALSE

8. Groundwater is a non-renewable resource.

TRUE / FALSE

Part C · Short Answer

2 questions × 3 points = 6 points

9. Explain the difference between *porosity* and *permeability*, and why an aquifer needs both.

10. Describe two problems that result from pumping more water out of an aquifer than is replaced.

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